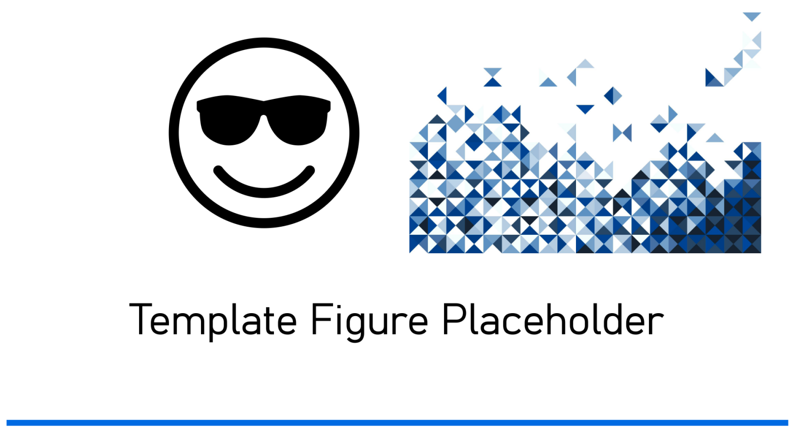
Graphical Abstract

Informational Content Analysis of Lost Languages

Anamaria Berea, Steven Esposito, Richard Witt



Highlights

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- Bullet 1 TODO:bullets
- Bullet 2
- Bullet 3

Informational Content Analysis of Lost Languages

Anamaria Berea[a], Steven Esposito[b], Richard Witt[c,*]

[a]*Computational Social Sciences, Associate Professor, George Mason University, 4400 University Drive, Fairfax, 22030, Virginia, USA*
[b]*Computational and Data Sciences, PhD Student,George Mason University, 4400 University Drive, Fairfax, 22030, Virginia, USA*
[c]*Computational and Data Sciences, PhD Student,George Mason University, 4400 University Drive, Fairfax, 22030, Virginia, USA*

Abstract

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PI GUIDANCE:

Frame the paper as a comparative analysis between ML specific methods (HDBSCAN) and information theoretic (and statistical) methods and compare.

It doesn't matter if these languages are logographic or syllabic, or alien. What I want you to focus on is on what kind of information we can extract from data that we don't know too much about (is agnostic). Assume these are dolphin languages or alien languages.

JOURNAL GUIDANCE:

Research Reports describe the results of original research of exceptional importance. The preferred length of these articles is 6 pages, but PNAS Nexus allows articles up to a maximum of 12 pages. A standard 6-page article is approximately 4,000 words, 50 references, and 4 medium-size graphical elements (i.e., figures and tables).

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This study investigates what information can be recovered from symbolic data such as lost languages or as yet undiscovered scripts. It compares two exploratory approaches that require no prior knowledge of the data or language. Machine learning clustering and information theoretic measurements. HDBSCAN is used to uncover recurring groups without predefined

*Corresponding author
Email address: `rwitt4@gmu.edu` (Richard Witt)

categories. Information theoretic measures quantify redundancy, uncertainty, and unicity distance as well as applying embeddings methods.

Keywords: TODO: Keywords Keyword1, keyword2, keyword3
TODO: Update for new article
PACS: 01.78.+p, 89.20.Dd, 01.75.+m, 89.70.Cf
TODO: Update for new article
2000 MSC: 68T50, 91B82, 62P12, 91F99

1. Introduction

Understanding what information is recoverable from unknown symbolic system would increase knowledge and understanding in archeology, anthropology, history, and linguistics. Speculatively, methods used to understand lost languages or scripts could also support the initial analysis of symbolic systems associated with a hypothetical extratresserial technological artifact, such as a plaque-like inscription on a spacecraft []. Research has been unable to yield satisfactory reproducible decipherments of lost scripts and basic questions such as Rongorongo's status as a writing system is still an open question[1]. Our Barthel created the now canonical transliterations for Rongorongo in 1958 [2]. Transliterations for Linear A and Indus Valley were provided by TODO in TODO [3] and TODO in TODO [4] respectively. Over six decades of research has been unable to provide scientifically verifiable, or repeatable interpretations or decipherments of these languages.

This paper attempts to apply Natural Language Processing, Machine Learning, and information-theoretical approaches to determine if any approaches may yield favorable insight into completely unknown languages. We first determine if there is enough informational content in the corpora for theoretical decipherment to be possible using information-theoretical techniques. We then apply machine learning techniques such as HDBSCAN categorization and embeddings to uncover available insight.

The remainder of this paper is structured as follows: Section 2 describes initial and current efforts to decipher the lost scripts. Section 3 details the source transliterations used in the analysis. Section 4 describes the methodology and rationale used to determine decipherability and writing system. Section 5 provides the analysis and results. Finally Section 6 discusses the results and describes potential paths for future work.

2. Literature Review

2.1. Easter Island and Rongo Rongo

Efforts to translate and decipher have spanned centuries. Rapa Nui is still the spoken language of the indiginous people of Easter Island. []

Father Sebastian Englebert creates Rapa Nui to Spanish Dictionary in 1938, translated to spanish in and updated in 1978.[5]

Rapa Nui to Spanish [5]

In his 1958 work *Grundlagen zur Entzifferung der Osterinselschrift* Barthel created the alphanumeric cataloging system and transliterations from glyph to alpha numeric characters [2].

Horley provides corrected glyphs and

Jacques Guy provided the tablet detail [6] and original web resources Philip Spaelti used to create Kohaumotu.org which provides the glyph library this paper uses as input data. [7]

2.2. Linear A

3. Data

Two distinct data sets were used for this paper. One for Rongo Rongo and one for Linear A. Rongo Rongo is the extinct script of Easter Island [8]. Linear A is also a lost script that was used on Crete during the second millennium BCE [3].

3.1. Rongo Rongo

Data available at the Rongorongo corpus (codes only)[7] was used for all analysis of Rongo Rongo. This site relies on the work initially recorded by Barthel in 1958 [2] which as been assembled in to an impressive interactive website greatly aiding in the understanding of Rongo Rongo glyphs. This paper relies on the accuracy of transliterations from this prior work.

The Rongo Rongo corpus consists of 26 Tablets Coded A though Z and retrieved from the Kohaumotu Rongorongo codes page [7]. The updated 3 character glyph codes were used to create an encoding library of all encodings. Instead of using the glyphs themselves we rely directly on the index created by Barthel to represent a manual encoding of glyphs [2].

The glyphs are numbered in a systematic manner. Each row is intended to categorize the glyphs with Row 0 containing the simplest forms while higher rows increase in complexity. While only 90 rows are shown 799 total rows were categorized by Barthel and avaliable in [2] and in the Kohaumotu.org website. [7].

To create the index used in this analysis the row and column are concatenated to result in the three digit index used. In this manner 000 represents the blank space in the upper left. and 001 represents both variants

Figure 1: **Rongo Rongo Great Washington Tablet.** Images created by the Smithsonian and compliled by .

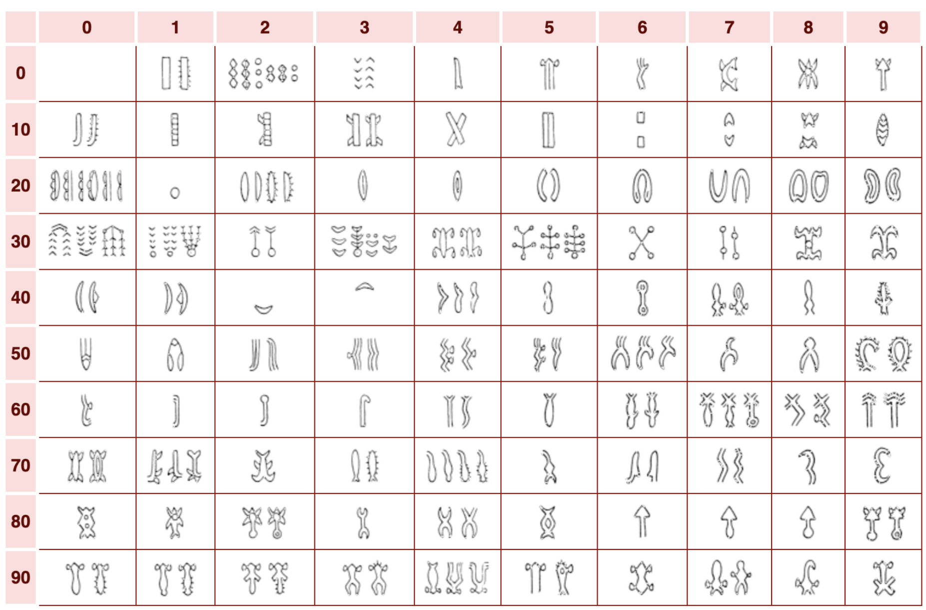

Figure 2: **Rongo Rongo Glyph Examples.** TODO.

Written in serpentine order sequenced in this order. TODO: consider left to right and right to left for both. does it matter? Could just say we assume temporality and were able to observe the message. this would not be the case in a alien society reading our space craft plates.

3.2. Linear A

A logosyllabic writing system [3] Written left to right, but some right to left and sometimes boustrophedon. In boustrophedon alternate lines of writing are reversed similar in directionality as an ox turns whiel plowing. []

Emma Bennett created a signory and numbering system in 1964 []

The entire linear A corpus contains around 1400 inscriptions with 7400 symbols. An additional 1100 objects are too poorly preserved to be readable and are not counted as part of this corpus.[]

Most Linear A writing has been found on the island of Crete and the Aegean islands for place and religious writings. The artifacts were written on clay, stone, gold, jewelry, and pottery. []

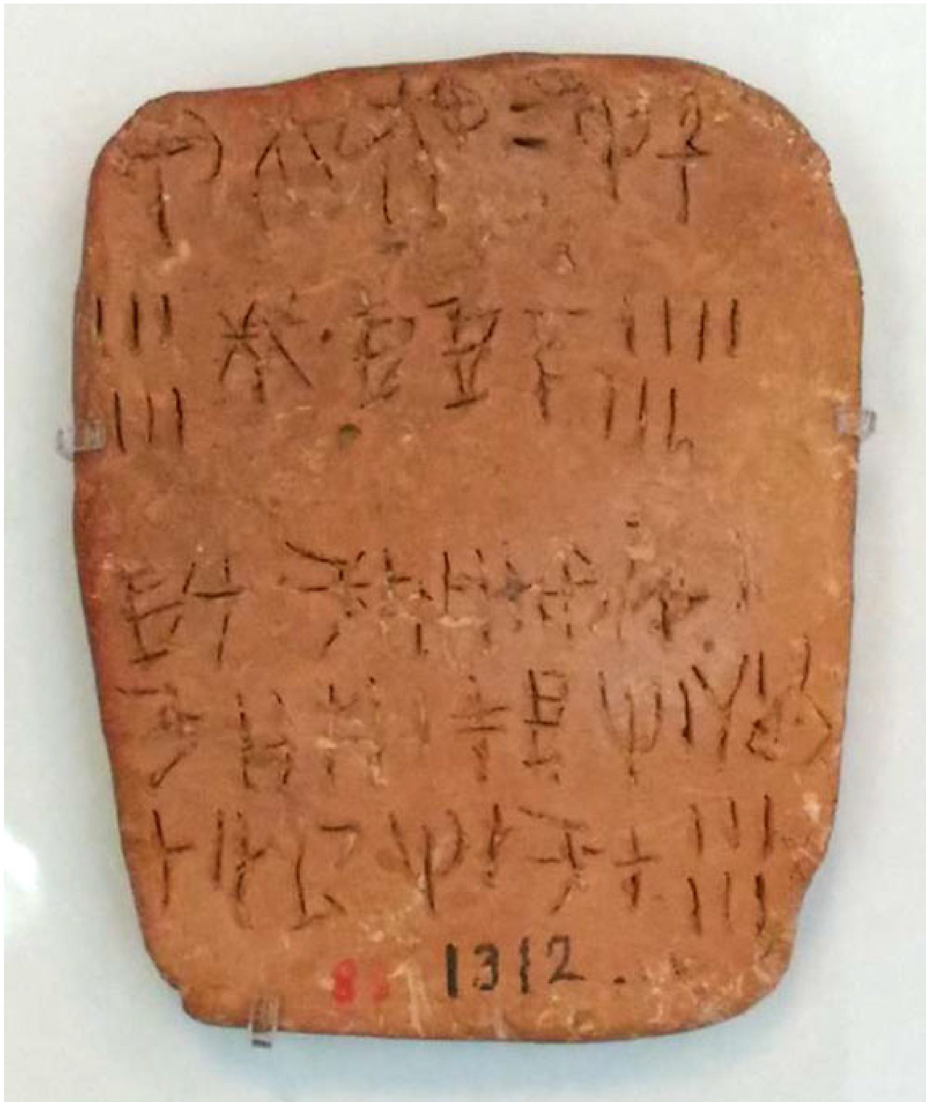

Figure 3: **Minoan clay tablet.** Clay tablet on display at the Archaeological Museum of Heraklion inscribed with Linear A. Image credit: Ester Salgarella.

The data used in this paper was sourced directly from Robert Hogan's Linear A Explorer which is a visulaizaiton tool for exploring the Linear A Language. [9] https://lineara.xyz/

The Linear A Explorer uses images and transcriptions from Luis Godart and Jean-Pierre Olivier [3] as well as the tabulations by George Douros. [9]

[]

4. Methodology and Rationale

We apply a machine learning clustering pipeline centered around an HDBSCAN model and information-theoretic techniques as complementary methods. HDBSCAN attempts to group data points into separable clusters by evaluating multivariate similarity among informational features. HDBSCAN is especially useful in unknown languages since it creates separation without requiring labels or a predetermined number of clusters. Information-theoretic measures summarize corpus properties such as uncertainty, redundancy, and properties relevant to decipherability. Shannon entropy measure the uncertainty in the distribution while redundancy reflects predictable or repeatable structure [10]. Both approaches distinguish patterns that can be inferred from the data without linguistic or archeological interpretations.

Data transformation techniques used in this paper require the symbols of both the Rongorongo and Linear A scripts to be analyzed as individual symbols (uni-grams) and sequences of symbols (n-grams). When a technique or data processing stage uses both uni-grams and n-grams we refer to them collectively as observations.

4.1. Data Preparation

4.1.1. Tokenization

*** NOT IN CODE YET *** Each individual symbol was tokenized by simple indexing. Each visibly atomic symbol identified by prior research such as [2] [3] [4] is assumed to be the atomic element of information or symbol. Each symbol is simply a distinct indexed item such as 001 for Rongorongo or *** INSERT COOL glyph *** in the case of Linear A.

4.1.2. Sequence Construction

TODO: Reference why this is used in cryptoanalysis and why it might apply here. Each corpus is sequenced as one long single sequence. No boundary or gap tokens are used as these would require an assumption of knowledge on the underlying structure of the communication. This places empty spaces on the same informational level as symbols. Let the resulting ordered token sequence be

$$X = (x_1, x_2, \dots, x_L),$$

where L is the number of tokens in the corpus.

4.1.3. Frequency Rank Normalization

*** NOT IN CODE YET *** Each corpus is frequency rank normalized independently before the corpora are merged. Let $\mathcal{S}_c$ denote the set of observed symbols in corpus c, with corpus frequency $f_c(s)$ calculated. Signs are then ordered from most to least frequent. Tied frequencies are assigned their mean rank.

Since the corpora contain different numbers of symbols, ranks are normalized with:

$$\rho_c(s) = \frac{r_c(s) - 1}{|\mathcal{S}_c| - 1},$$

where $r_c(s)$ is the frequency rank of symbol $|\mathcal{S}_c|$. $\rho_c(s) = 0$ represents the most frequent sign, while values approaching 1 represent increasingly rare signs.

Frequency rank normalization removes sign identity and gives the two scripts a shared feature space. This shared feature space allows HDBSCAN to compare patterns of relative symbol frequency and prevents the algorithm from separating the corpora merely based on different symbol inventories.

4.2. Machine Learning with HDBSCAN

HDBSCAN pipeline consists of seven separate stages. Corpus Ingestion, TF-IDF Vectorization and UMAP Dimensionality Reduction prepare the data for the HDBSCAN algorithm. The HDBSCAN algorithm analyzes the data to form groups by evaluating the relative density and proximity of the data points. Topic extraction supports interpretation of resulting clusters and 3D Projection prepares the HDBSCAN output for 3 dimensional visualization. The following paragraphs describe each step of the process.

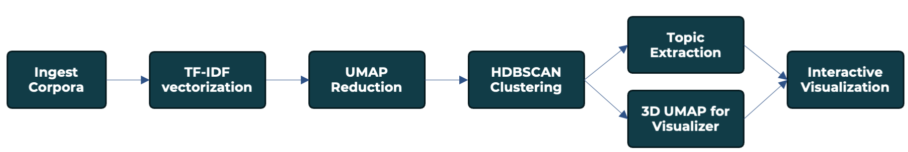

Figure 4: **HDBSCAN Pipeline.** HDBSCAN pipeline showing Corpus Ingestion, TF-IDF Vectorization, UMAP Dimensionality Reduction, HDBSCAN Clustering, Topic Extraction, Three Dimensional Projection, and Interactive Visualization used in the analysis.

4.2.1. Corpus Ingestion

Rongorongo and Linear A transcriptions are ingested into the pipeline and merged into a single document collection. Source identities are kept for subsequent evaluation and visualization but are not used to guide clustering or other algorithms.

4.2.2. Vectorization

Each observed symbol is converted into a vector based on uni-gram and bi-gram features using TF-IDF. Vectorization was restricted to bi-grams due to the relatively low amount of data available in order to ensure *** JUSTIFY it kept erroring somehow ***.. TF-IDF emphasizes terms that are important in the sequence of text yet relatively uncommon in the corpus [11].

4.2.3. Dimensioality Reduction

Dimensionality reduction was performed using the UMAP algorithm. This reduces the high dimensionality resulting from the TF-IDF vectorization into 42 dimensions *** JUSTIFY ***. Cosine distance was chosen because it compensates for document size by length normalizing the vectors to unit vectors [11].

4.2.4. HDBSCAN Clustering

Clustering is performed by HDBSCAN. The algorithm searches the 42-dimensional space for stable regions of high density. HDBSCAN was chosen because it does not require the number of clusters to be selected before hand. A minimum cluster size of 50 *** JUSTIFY *** was used. Once the model identifies the stable regions of high density is assigns each symbol or sequence of symbols to a stable cluster. If the observation is not a fit for one of the clusters it is labeled as noise.

4.2.5. Topic Extraction

Each cluster is then labeled with a topic using the five terms with the greatest TF-IDF weight among it's observations. This helps identify the symbols or sequences of symbols best represent the cluster.

4.2.6. 3D Projection

In order to display the 42 dimensions used to cluster the symbols in human readable form UMAP was once again used for dimensionality reduction. The 42 dimensions projected onto a 3-dimensional space superimpose the higher dimensional space. As a consequence the displayed coordinates are illustrative and are not the actual space used for clustering.

4.2.7. Interactive Visualization

The visualization shown is a static representation of a interactive visualization that can be used to display the observations by cluster, characteristic terms, and a preview of the corresponding observation.

4.3. Information Theoretical

The informaiton theoretical pipline starts with raw data as defined in 3. n-grams of orders 1-4 ***JUSTIFY*** are prepared from this raw data. Each data set is analyzed from an frequency,

4.3.1. N-gram Construction

Overlapping n-grams were generated for $n \in \{1, 2, 3, 4\}$. The i-th n-gram is defined as

$$g_i^{(n)} = (x_i, x_{i+1}, \ldots, x_{i+n-1})$$

where where x_i denotes the token at position i in a sequence of length L, and $i \in \{1, \ldots, L-n+1\}$.

The tokens in each n-gram were joined with hyphens providing a serialized representation of each sequence.

4.3.2. Frequency estimation

Post n-gram construction frequency estimation is once again employed. For each value of n, the the count, c, of every distinct n-gram is calculated. If c_j is the frequency of the j-th distinct n-gram then its probability of occurrence is estimated with

$$p_j = \frac{c_j}{T_n}.$$

Where T_n is the total number of all n-grams in the set.

If we let K_n is the number of distinct n-grams then corpus diversity can be summarized calculating the proportion of unique observations for each $n - gram$ set

$$Q_n = \frac{K_n}{T_n}.$$

resulting in a set of proportions between 0 and 1 with higher values of Q_n indicating more rare occurrences of that $n - gram$ sequence and lower values indicating higher repetition.

4.3.3. Entropy and Redundancy

The Shannon entropy of each $n - gram$ set was calculated in bits using methods outlined in [12]

$$H_n = -\sum_{j=1}^{K_n} p_j \log_2 p_j.$$

The max potential entropy in the $n - gram$ set is defined as the entropy of a uniform distribution containing the same number of distinct symbols:

$$H_{n,\max} = \log_2 K_n.$$

Normalized redundancy was calculated by first calculating absolute redundancy:

$$D_n = H_{n,\max} - H_n,$$

and dividing by the maximum possible redundancy:

$$R_n(D_n) = \frac{D_n}{H_{n,\max}}$$

In this way D_n measures the redundancy as departures from uniformity in bits per n-gram and R_n shows the same departure from uniformity as a proportion of maximum entropy following the work of [10].

4.3.4. Keyspace, Unicity Distance and Theoretical Decipherability Margin

Possible key space and unicity distance are calculated to determine if the set of n-grams contains enough information to make a theoretical decipherment possible.

A key space is the set of all possible keys in a cypher model. In this model the key space is all possible permutations of the $n - gram$ sets and is calculated

$$|\mathcal{K}_n| = K_n!,$$

and the corresponding key-space size in bits is

$$B_n = \log_2(K_n!).$$

Observed unicity distance for the $n - gram$ set is then calculated by

$$U_n = \frac{B_n}{D_n}.$$

The resulting values from the unicity distance calculation estimates the number of $n{-}gram$ observations required for redundancy to overcome uncertainty in the substitution key and contain enough information to identify a unique mapping [12].

Finally the theoretical decipherability margin was calculated by subtracting the unicity distance U_n from the total number of $n - grams$ T_n in the set. A positive margin indicates the corpus, as sequenced $n - gram$ sets, is estimated to contain sufficient information to identify the key as unique. It therefore indicates informational-theoretic decipherability. It does not guarantee the key can be recovered computationally, but that it is possible to be accomplished given the assumptions inherent in the data preparation.

5. Analysis and Results

TODO

6. Discussion and Future Work

The methods in this paper cannot decipher the meaning or linguistic composition of an unknown symbolic system. Instead they determine if the available data contain non-random regularities and what scale those regularities happen. It can also be used to determine if the corpus is of sufficient size and non-random enough to support more specific analysis.

May be worth attempting to read un-readable Linear A scripts since the only $1 - gram$ set has enough information to be theoretically decipherable. Since the Linear A script is known to have complex symbols combined via ligature [3]

it is likely the true decipherablility requires more information than the $1 - gram$ sequense implies.

Appendix A. TODO

Year	Title	TODO
2001	Astronomy and Astrophysics for the New Millennium	Astrophysics
2003	The Sun to the Earth—and Beyond: A Decadal Research Strategy in Solar and Space Physics	Heliophysics
2003	New Frontiers in the Solar System: An Integrated Exploration Strategy	Planetary Science
2007	Earth Science and Applications from Space: National Imperatives for the Next Decade and Beyond	Earth Science

Table A.1: Decadal Surveys Used in Analysis [13]

Appendix B. TODO2

This section details TODO

TODO1: Follow a `yyyy-DS-Category-####.pdf` format where `####` is the National Academies document number. `DS` is used to categorize all Decadal Surveys. `MA` replaces `DS` for Midterm Assessments, and `SP` is used for the Special Publication in 2015 that identified lessons learned.

TODO2: Follow a `yyyy_SSP_Nasa_File_Name.pdf` format. `SSP` represents Science mission directorate Science Plans. This is replaced with `AD` to indicate Additional Documentation, `NSP` to indicate NASA Science Plan, or `AD` to indicate additional documentation present on the SMD website.

References

[1] F. Tamburini, On automatic decipherment of lost ancient scripts relying on combinatorial optimisation and coupled simulated annealing, Frontiers in Artificial Intelligence 8 (2025). doi:10.3389/frai.2025.1581129.

[2] T. S. Barthel, Grundlagen zur Entzifferung der Osterinselschrift, Cram, de Gruyter, Hamburg, 1958.

[3] E. Salgarella, S. Castellan, Sigla: The signs of linear a, a palaeographical database (2021).

[4] N. Yadav, H. Joglekar, R. P. N. Rao et al., Statistical analysis of the indus script using n-grams (2010). arXiv:0901.3017.
URL https://arxiv.org/abs/0901.3017

[5] S. Englert, Diccionario Rapanui-Español, 1948, translated from Spanish to English by Jacques Guy.

[6] J. Guy, Rongorongo or the hieroglyphs of the easter island tablets, archived and mirrored by Philip Spaelti (2005).
URL http://kohaumotu.org/rongorongo_org/

[7] P. Spaelti, Glyph library.
URL http://kohaumotu.org/Rongorongo/Glyph_Library/index.html

[8] S. R. Fischer, Rongorongo: The Easter Island Script: History, Traditions, Text, Oxford University Press, Oxford, 1997.

[9] J. R. Karajgikar, A. Al-Khulaidy, A. Berea, Computational pattern recognition in linear a (2025).

[10] C. E. Shannon, A mathematical theory of cryptography, Confidential report, Bell Telephone Laboratories (1945).

[11] C. D. Manning, P. Raghavan, H. Schütze, Introduction to Information Retrieval, Cambridge University Press, 2008, see the vector space model discussion of cosine similarity as length-normalized vector comparison.
URL https://nlp.stanford.edu/IR-book/

[12] C. E. Shannon, Communication theory of secrecy systems, Bell System Technical Journal 28 (4) (1949) 656–715. doi:10.1002/j.1538-7305.1949.tb00928.x.

[13] National Academies of Sciences, Engineering, and Medicine, Decadal surveys, National Academies Press, (Accessed: 3 December 2024).
URL https://nap.nationalacademies.org